



Topic Flow

LangGraph-powered research signal tracker and science communication assistant for medical use cases.

Streamlit · OpenAlex · SBERT · BERTopic · LLM · LangGraph

Demo

huggingface.co/spaces/carlosocantod/GOSIM-agent

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Medical question

“What are the latest treatments for malaria?”



**Emerging topics
+ social media draft**

Live demo: Hugging Face Spaces

The problem

Clinicians and medical researchers cannot manually keep pace with biomedical literature.

1.5M+

biomedical papers published each year

4,000+

papers per day to scan, interpret, and translate

Hours → minutes

goal: compress literature scanning and writing into one query



Literature overload



Signal detection



Science communication

TopicFlow automates the scan, compares recent research against a baseline, and turns findings into audience-specific content.

What I built

A Streamlit app that turns a plain-language medical question into research signals and communication drafts.

1. Medical query analysis

Extracts canonical biomedical keywords and rejects non-medical queries.

2. Literature signal discovery

Finds dominant topics in recent papers and compares them with a six-month baseline.

3. Ready-to-publish drafts

Generates content for LinkedIn, Instagram carousel, or Twitter/X thread.

4. Deployment on HuggingFace Space

Signal workflow

What the LLM-enhanced topic model checked during this run.

✓ Medical query detected.
Keywords: malaria treatment, antimalarial therapy, antimalarial drugs

✓ Created a literature scan plan.

✓ Selected current and baseline literature windows.

✓ Retrieved 500 current-period papers; kept 200 with abstracts after reranking.

✓ Search quality check: good
— The sample documents are predominantly relevant to malaria treatments and include recent advances, with sufficient abstracts fetched.

✓ Modeled current literature into 6 relevant topics.

✓ Retrieved 500 baseline-period papers; kept 200 with



LangGraph-powered deep research and science communication agent for medical use cases — built on an LLM-enhanced topic model.

Enter your medical research query

Which are the latest treatments for malaria?

Explore literature signals

Keywords used: malaria treatment, antimalarial therapy, antimalarial drugs

> Literature signal plan

> Search quality check

Current literature signals: Feb 2026 – May 2026

120 papers across 6 topics

> Malaria Surveillance and Elimination Strategies — 49 papers

> Antimalarial Drug Evaluation and Cost Analysis — 24 papers

> Malaria Vaccine Development and Immune Correlates — 18 papers

> Plasmodium falciparum Biology and Immune Evasion — 17 papers

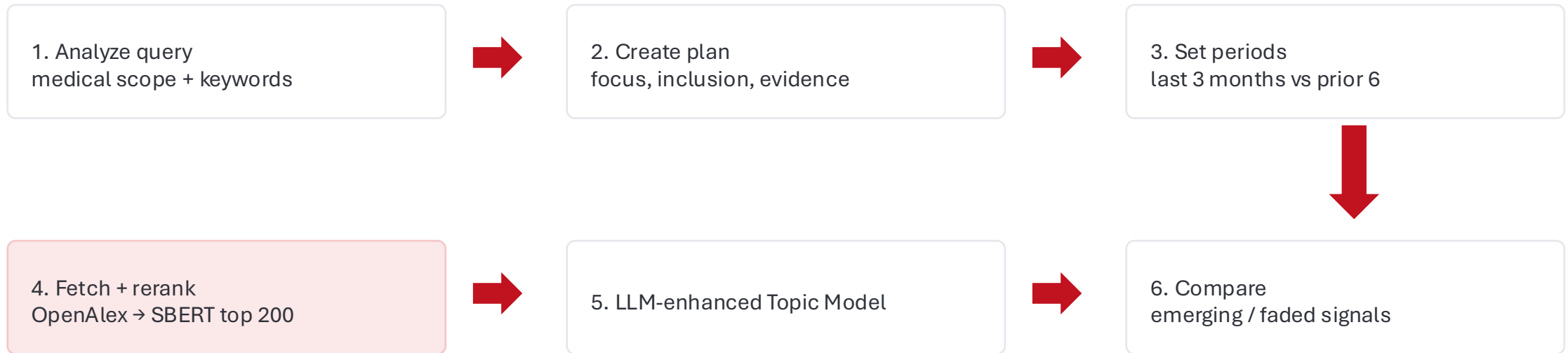
> Malaria in Pregnancy and Antenatal Care — 10 papers

> Community-Based Malaria Control Experiences — 2 papers

Design principle: expensive topic modeling runs once; audience and platform drafts can be regenerated independently.

Core LangGraph pipeline

The research workflow is decomposed into auditable nodes with conditional search refinement.



Cached core result

The core result is reused for multiple audiences and platforms, avoiding unnecessary reruns of paper retrieval and topic modeling.

Implementation highlights

The project combines reliable literature retrieval, deterministic topic discovery, and LLM-based interpretation.

OpenAlex API

Queries recent biomedical papers by keyword, publication date, English language, DOI, and abstract availability.

SBERT reranking

Embeds abstracts and keeps the most semantically relevant papers for the original query.

BERTopic + KMeans

Clusters papers deterministically without UMAP, making results more repeatable and stable.

LLM topic cleanup

Merges near-duplicate topics, filters irrelevant clusters, and summarizes each topic. Deals with open question in NLP on how to select number of topics and clean inference.

Period comparison

Compares current topics with a prior baseline to identify emerging and faded signals.

Streamlit + caching

Keeps the UI responsive while preventing expensive reruns.

Result: a reproducible medical-literature signal tracker with interpretable output rather than a black-box search summary.

Science communication layer

After the core research pass, the app drafts content for the selected audience and platform.

Audience selector

Clinicians

Patients

Journalists

Researchers

Platform selector

LinkedIn post
Professional narrative

Instagram carousel
5–6 slide story

Twitter/X thread
4–6 concise posts

The communication layer also suggests follow-up questions, including questions about uncertainty and evidence strength.

What changed vs. Aug 2025 – Jan 2026?

The research focus shifted from broad themes of disease burden, antimicrobial resistance crisis, and health system resilience to targeted malaria elimination strategies including surveillance, drug evaluation, and pregnancy interventions. Basic science on *Plasmodium falciparum* survival mechanisms emerged as a new area, while vaccine research pivoted from equity and manufacturing to development and cold chain. Topics on medicinal plants and general AMR disappeared, replaced by vector-borne disease control integrating resistance concerns. Community-based malaria control persisted, but health system adaptations narrowed in scope.

Emerging signals: Antimalarial Drug and Diagnostic Evaluation, Malaria in Pregnancy Interventions, Malaria Surveillance and Elimination, *Plasmodium falciparum* Survival Mechanisms, Vector-Borne Disease Control and Resistance, Vaccine Development and Cold Chain

Less prominent than baseline: Global Vaccine Equity and Manufacturing, Antimicrobial Resistance Crisis, Medicinal Plants for Infectious Diseases, *Plasmodium* Drug Resistance Mutations, Health System Resilience and Adaptation, Malaria Burden in Vulnerable Populations

> Baseline topics: Aug 2025 – Jan 2026

AudiencePlatform

General publicInstagram

✓ Drafting Instagram content...

LinkedInInstagramTwitter/X

Instagram draft

The Latest in Malaria Treatment

Carousel slides

Slide 1 — A New Wave of Hope

Malaria still threatens millions, but breakthrough treatments are on the way. Science is delivering powerful new tools to fight this deadly disease.

Slide 2 — Smarter Drug Testing

Scientists are using advanced bioassays to measure drug potency. This helps identify the most effective antimalarials to combat resistance.

Slide 3 — Vaccines in the Pipeline

Malaria vaccine trials are showing strong immune responses. These shots could protect those most at risk, like children in endemic areas.

Slide 4 — Better Detection, Faster Care

Outcome and next steps

A deployed demo that demonstrates an end-to-end research-to-communication workflow.

Delivered

- Streamlit UI
- Hugging Face Spaces demo
- Docker deployment
- GitHub Actions auto-sync
- OpenAI-compatible LLM support

Value

- Less manual literature triage
- Clear emerging/faded signal narrative
- Reusable core research result
- Drafts tailored to audience and platform

Future improvements

- Follow-up chat over the results
- LangGraph / LangSmith tracing
- Topic timelines and charts
- PDF / JSON export
- PubMed, bioRxiv, clinical trial sources

Demo

<https://huggingface.co/spaces/carlosocantod/GOSIM-agent>

Takeaway: TopicFlow shows how agentic workflows can make biomedical research signals easier to find, compare, and communicate.

Thank you very much for your attention!

Hope you enjoy TopicFlow 🙏

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